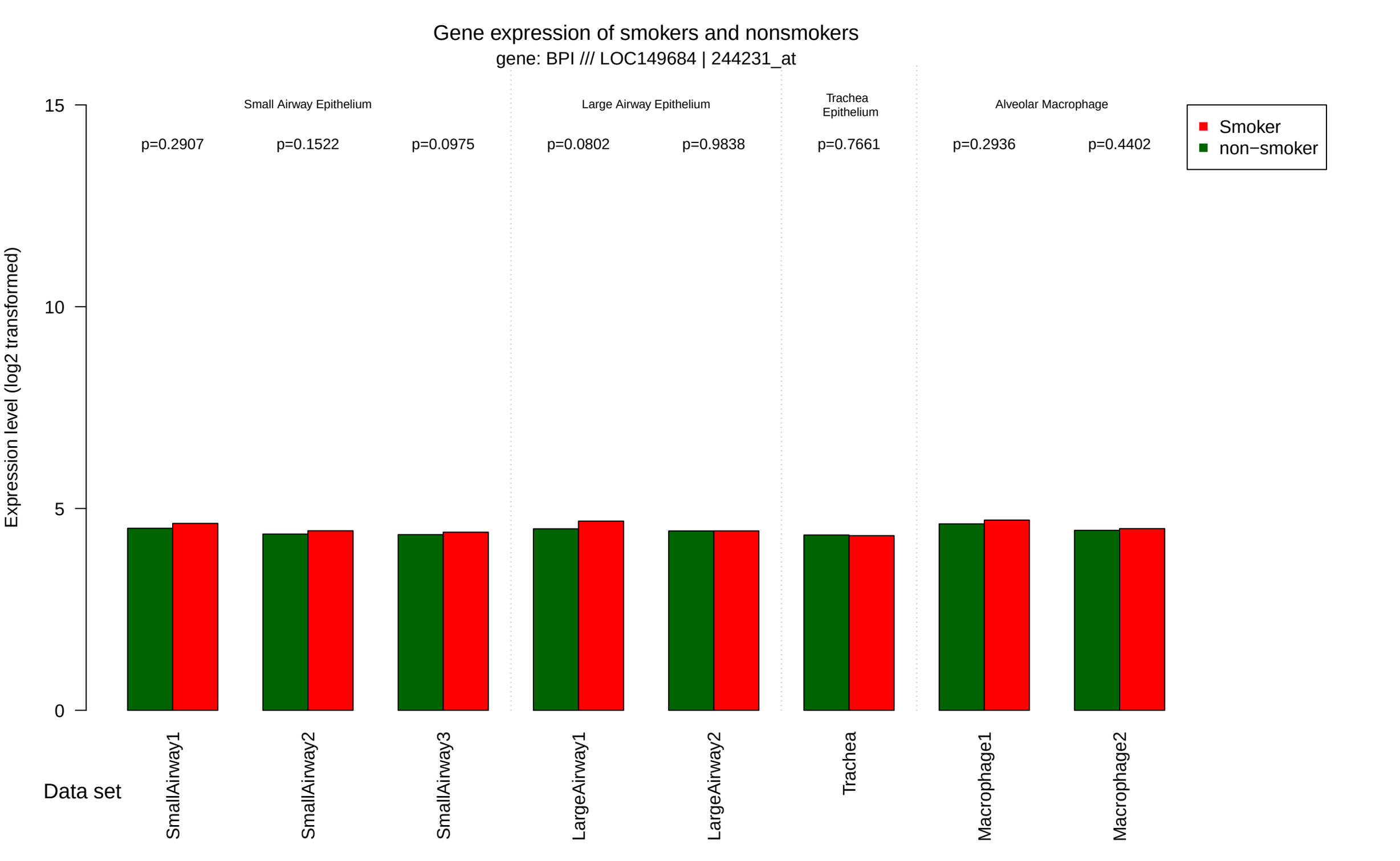
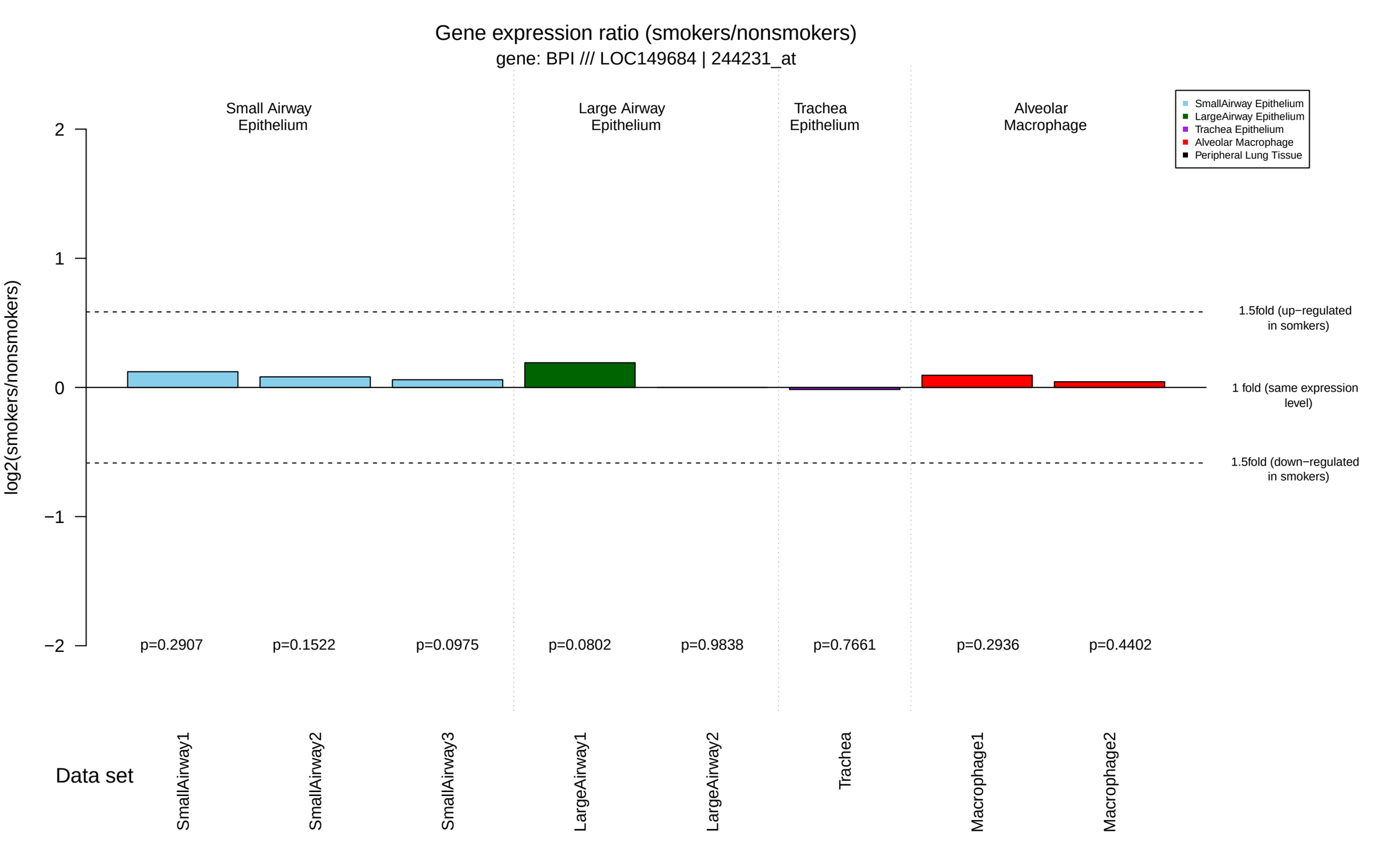
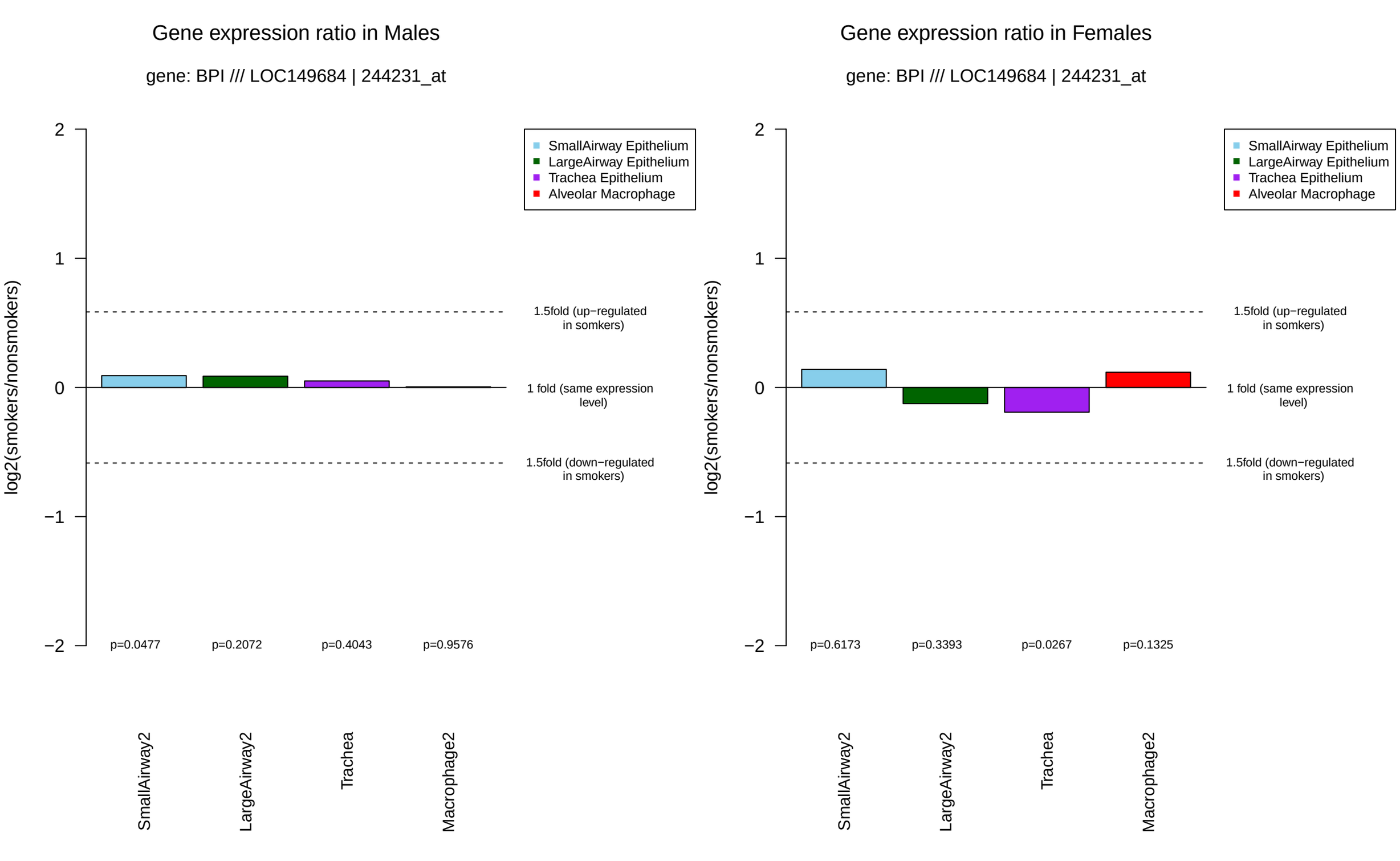
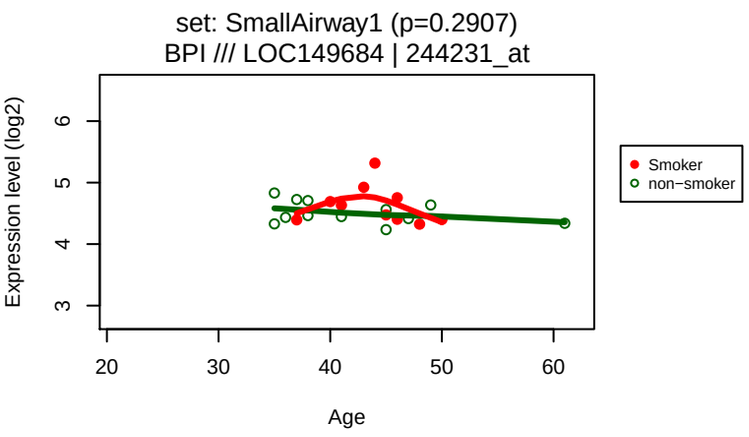
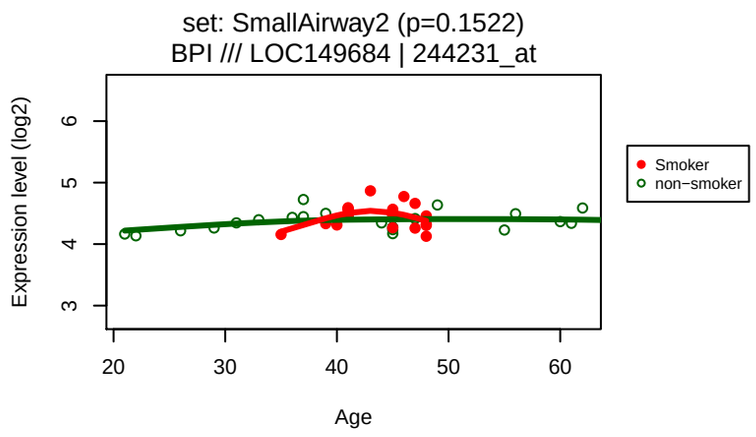
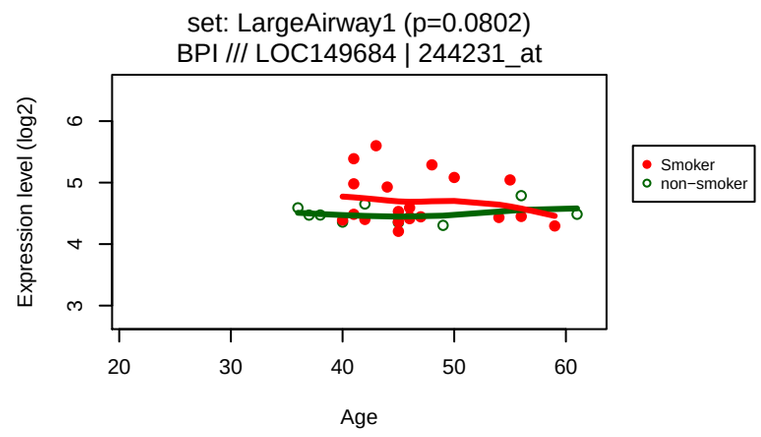

Gene expression of smokers and nonsmokers
gene: BPI /// LOC149684 | 244231_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2907
p=0.1522
p=0.0975
p=0.0802
p=0.9838
p=0.7661
p=0.2936
p=0.4402
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

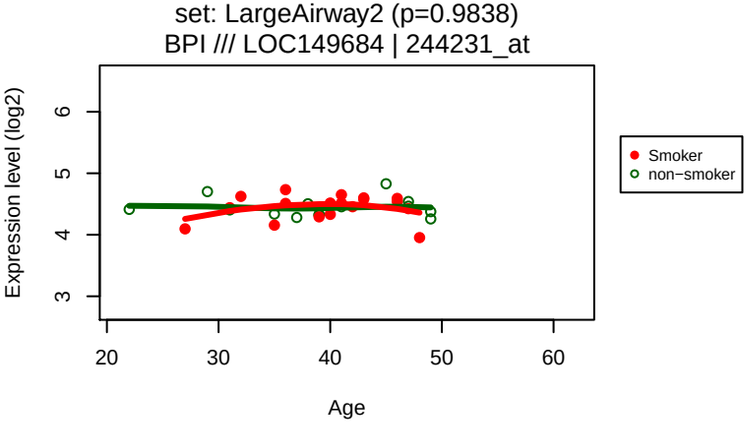

Gene expression ratio (smokers/nonsmokers)
gene: BPI /// LOC149684 | 244231_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2907
p=0.1522
p=0.0975
p=0.0802
p=0.9838
p=0.7661
p=0.2936
p=0.4402
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

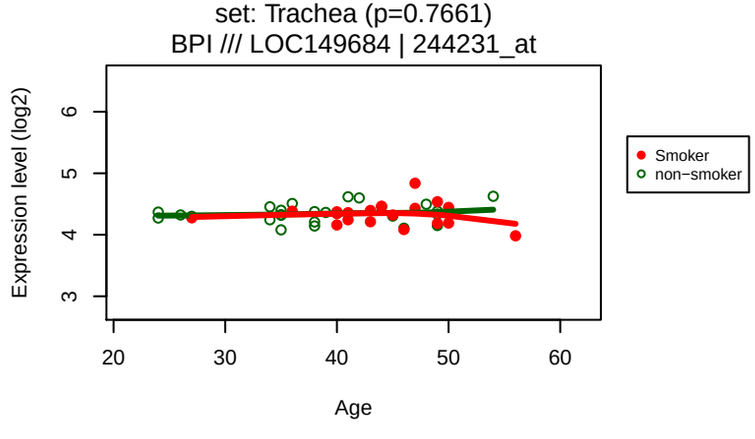

Gene expression ratio in Males
gene: BPI /// LOC149684 | 244231_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0477
p=0.2072
p=0.4043
p=0.9576
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: BPI /// LOC149684 | 244231_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6173
p=0.3393
p=0.0267
p=0.1325
SmallAirway2
LargeAirway2
Trachea
Macrophage2

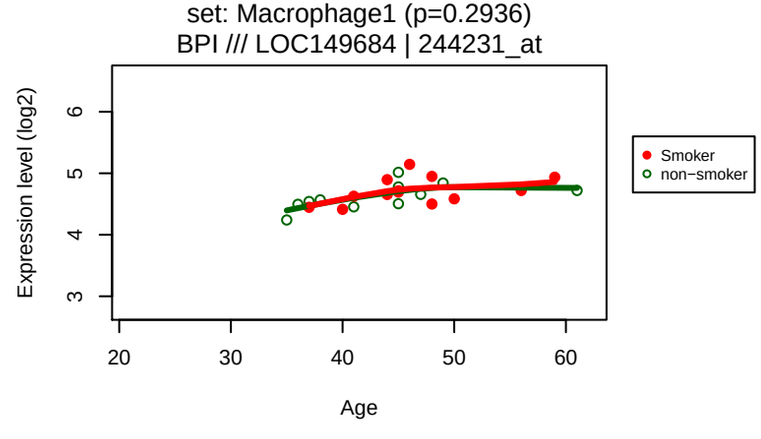

set: SmallAirway1 (p=0.2907)
BPI /// LOC149684 | 244231_at
Expression level (log2)
Age
Smoker
non−smoker

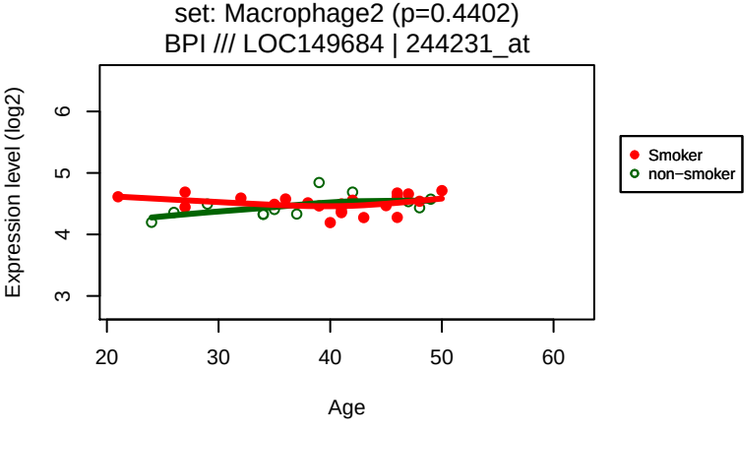

set: SmallAirway2 (p=0.1522)
BPI /// LOC149684 | 244231_at
Expression level (log2)
Age
Smoker
non−smoker

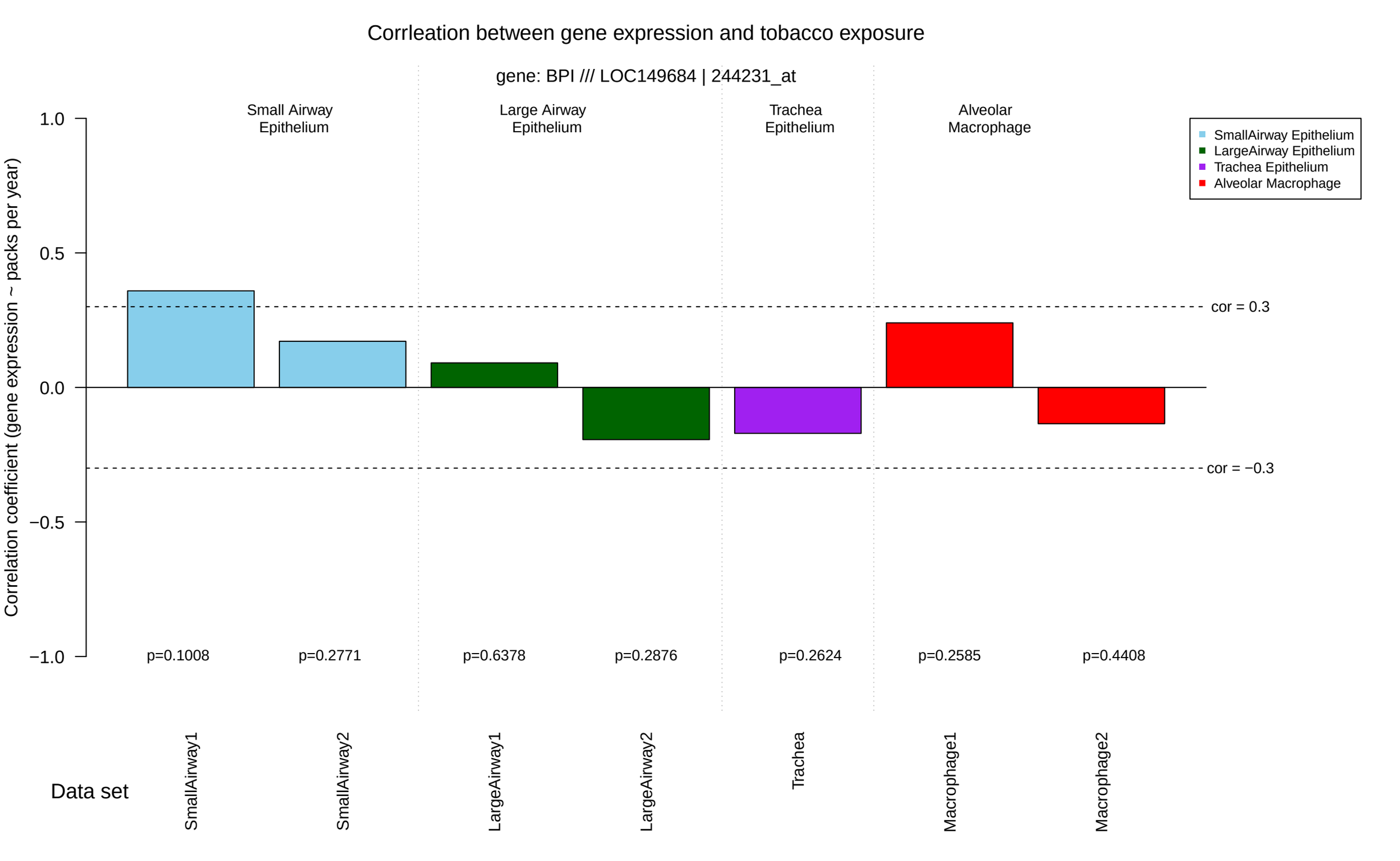

set: LargeAirway1 (p=0.0802)
BPI /// LOC149684 | 244231_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9838)
BPI /// LOC149684 | 244231_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7661)
BPI /// LOC149684 | 244231_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2936)
BPI /// LOC149684 | 244231_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4402)
BPI /// LOC149684 | 244231_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: BPI /// LOC149684 | 244231_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1008
p=0.2771
p=0.6378
p=0.2876
p=0.2624
p=0.2585
p=0.4408
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2